



BUBT | BANGLADESH UNIVERSITY OF
BUSINESS AND TECHNOLOGY

COMPUTER SCIENCE AND ENGINEERING

LAB REPORT

MID-TERM

Compiler Design (CSE-323)

Submitted By:

Kaniz Fatema

20245103154

Intake 53 (03)

Submitted To:

Md. Nahid Hossain

Lecturer

Dept. of CSE

July 07, 2026

Contents

1	Count Words from Sentence	3
1.0.1	Question	3
1.0.2	Solution	3
2	Identify Token	3
2.0.1	Question	3
2.0.2	Solution	3
3	Infix to Postfix and Prefix Conversion	6
3.0.1	Question	6
3.0.2	Solution	6
4	Identify Comment Line	9
4.0.1	Question	9
4.0.2	Solution	9
5	Identify Valid Keywords	10
5.0.1	Question	10
5.0.2	Solution	10
6	Recognize Valid and Invalid Identifier	11
6.0.1	Question	11
6.0.2	Solution	11
7	FIRST and FOLLOW Function	13
7.0.1	Question	13
7.0.2	Solution	13

1 Count Words from Sentence

1.0.1 Question

1. Write a LEX/C++ program to count words from a sentence.

1.0.2 Solution

```
1 #include <iostream>
2 #include <string>
3 using namespace std;
4
5 int main(){
6     string sentence; //My name is Kaniz
7     int Count = 1;
8
9     cout << "Enter a sentence: ";
10    getline(cin, sentence);
11
12    for(int i = 0; i<sentence.length(); i++){
13        if(sentence[i] == ' '){
14            Count++;
15        }
16    }
17    cout << "Total words: "<< Count <<endl;
18 }
```

Listing 1: C++ program to count words from a sentence.

Output

2 Identify Token

2.0.1 Question

1. Write a LEX/C++ program to identify tokens.

2.0.2 Solution

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS    GITLENS

Active code page: 65001

D:\code2026\cse324>cd "d:\code2026\cse324\lab-mid\" && g++ 01-count-words-f
from-sentence && "d:\code2026\cse324\lab-mid\"01-count-words-from-sentence
Enter a sentence: My name is Kaniz
Total words: 4

d:\code2026\cse324\lab-mid>
```

Figure 1: Output of word counting program

```
1 // Write a C++ program to identify tokens.
2 #include <iostream>
3 #include <string>
4 using namespace std;
5
6 int main()
7 {
8     string sentence, w = "";
9
10    cout << "Enter a statement: ";
11    getline(cin, sentence);
12
13    sentence += " ";
14
15    for (int i = 0; i < sentence.length(); i++)
16    {
17        if (sentence[i] != ' ')
18        {
19            w = w + sentence[i];
20        }
21        else if (w != "")
22        {
23            if (w == "int" || w == "float" ||
24                w == "char" || w == "double")
25            {
26                cout << "KEYWORD: " << w << endl;
27            }
28            else if (w == "+" || w == "-" ||
29                    w == "*" || w == "/" || w == "=")
30            {
31                cout << "OPERATOR: " << w << endl;
32            }
33        }
34    }
35 }
```

```

33     else if (w == ";" || w == "(" || w == ")" ||
34              w == "{" || w == "}")
35     {
36         cout << "SYMBOL: " << w << endl;
37     }
38     else if (w[0] >= '0' && w[0] <= '9')
39     {
40         cout << "NUMBER: " << w << endl;
41     }
42     else
43     {
44         cout << "IDENTIFIER: " << w << endl;
45     }
46     w = "";
47 }
48 }
49 }
50
51 return 0;
52 }

```

Listing 2: LEX/C++ program to identify tokens.

Output

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

Active code page: 65001

D:\code2026\cse324>cd "d:\code2026\cse324\lab-mid\" && g++ 02-identify-tokens.cpp
\code2026\cse324\lab-mid\'02-identify-tokens
Enter a statement: int a = 10
KEYWORD: int
IDENTIFIER: a
OPERATOR: =
NUMBER: 10

d:\code2026\cse324\lab-mid>

```

Figure 2: Output of token identification program

3 Infix to Postfix and Prefix Conversion

3.0.1 Question

1. Write a C++ program to convert an Infix to Postfix/Prefix expression using stack.

3.0.2 Solution

```
1 #include <cctype>
2 #include <iostream>
3 #include <stack>
4 #include <string>
5
6 using namespace std;
7
8 int precedence(char op)
9 {
10     if (op == '+' || op == '-')
11         return 1;
12
13     if (op == '*' || op == '/')
14         return 2;
15
16     return 0;
17 }
18
19 string infixToPostfix(string infix)
20 {
21     string answer = "";
22     stack<char> box;
23
24     for (char c : infix)
25     {
26         if (c == ' ')
27             continue;
28
29         if (isalnum(c))
30         {
31             answer += c;
32         }
33         else if (c == '(')
34         {
35             box.push(c);
36         }
37         else if (c == ')')
38         {
39             while (!box.empty() && box.top() != '(')
40             {
41                 answer += box.top();
```

```

42         box.pop();
43     }
44
45     if (!box.empty())
46         box.pop();
47 }
48 else
49 {
50     while (!box.empty() &&
51           box.top() != '(' &&
52           precedence(box.top()) >= precedence(c))
53     {
54         answer += box.top();
55         box.pop();
56     }
57
58     box.push(c);
59 }
60 }
61
62 while (!box.empty())
63 {
64     answer += box.top();
65     box.pop();
66 }
67
68 return answer;
69 }
70
71 string postfixToPrefix(string postfix)
72 {
73     stack<string> box;
74
75     for (char c : postfix)
76     {
77         if (isalnum(c))
78         {
79             box.push(string(1, c));
80         }
81         else
82         {
83             string right = box.top();
84             box.pop();
85
86             string left = box.top();
87             box.pop();
88
89             box.push(string(1, c) + left + right);
90         }
91     }

```

```

92
93     return box.top();
94 }
95
96 int main()
97 {
98     string infix;
99
100     cout << "Enter an infix expression: ";
101     cin >> infix;
102
103     string postfix = infixToPostfix(infix);
104     string prefix = postfixToPrefix(postfix);
105
106     cout << "Postfix: " << postfix << endl;
107     cout << "Prefix: " << prefix << endl;
108
109     return 0;
110 }

```

Listing 3: C++ program to convert an Infix expression to Postfix and Prefix using stack.

Output

The screenshot shows a terminal window with a dark background. At the top, there are tabs for 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL' (which is active), 'PORTS', and 'GITLENS'. Below the tabs, the text 'Active code page: 65001' is displayed. The terminal shows the following commands and output:

```

D:\code2026\cse324>cd "d:\code2026\cse324\lab-mid\" && g++ 03-infix-to-postfix-prefix.cpp
ix-prefix && "d:\code2026\cse324\lab-mid\"03-infix-to-postfix-prefix
Enter an infix expression: (A+B)*(C-D)/E
Postfix: AB+CD-*E/
Prefix: /*+AB-CDE

d:\code2026\cse324\lab-mid>

```

Figure 3: Output of Infix to Postfix and Prefix conversion program

4 Identify Comment Line

4.0.1 Question

1. Write a C/C++ program to identify whether a given line is a comment or not.

4.0.2 Solution

```
1 // Write a C/C++ program to identify whether a given line is a comment or not.
2 #include <iostream>
3 #include <string>
4 using namespace std;
5
6 int main(){
7     string a;
8     cout <<"Enter a string: "<<endl;
9     getline(cin, a);
10
11     int n = a.length();
12
13     if((a[0] == '/' && a[1] == '*') && (a[n-1] == '/' && a[n-2] == '*')){
14         cout << "This is a comment." <<endl;
15     }
16     else if((a[0] == '/' && a[1]== '/')){
17         cout << "This is a comment." <<endl;
18     }
19     else{
20         cout<<"This is not a comment."<<endl;
21     }
22
23 }
```

Listing 4: C/C++ program to identify whether a given line is a comment or not.

Output

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

Active code page: 65001

D:\code2026\cse324>cd "d:\code2026\cse324\lab-mid\" && g++ 04-check-comment-line.cpp
&& "d:\code2026\cse324\lab-mid\04-check-comment-line
Enter a string:
?who am I
This is not a comment.

d:\code2026\cse324\lab-mid>

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

Active code page: 65001

D:\code2026\cse324>cd "d:\code2026\cse324\lab-mid\" && g++ 04-check-comment-line.cpp
&& "d:\code2026\cse324\lab-mid\04-check-comment-line
Enter a string:
//I love my country Bangladesh
This is a comment.

d:\code2026\cse324\lab-mid>
```

Figure 4: Output of comment line identification program

5 Identify Valid Keywords

5.0.1 Question

1. Write a C/C++ program to identify valid keywords.

5.0.2 Solution

```
1 // Write a C/C++ program to identify valid keywords.
2 #include <iostream>
3 #include <string>
4
5 using namespace std;
6
7 int main() {
8     string word;
9     bool found = false;
10
11     string keywords[] = {
12         "int", "float", "char", "double", "if", "else",
13         "for", "while", "do", "return", "break", "continue",
14         "switch", "case", "void", "class", "struct"
15     };
16     int n = sizeof(keywords) / sizeof(keywords[0]);
17
18     cout << "Enter a word: ";
```

```

19     cin >> word;
20
21     for (int i = 0; i < n; i++) {
22         if (word == keywords[i]) {
23             found = true;
24             break;
25         }
26     }
27
28     if (found) {
29         cout << "This is a Keyword.";
30     } else {
31         cout << "This is not a Keyword.";
32     }
33 }
34

```

Listing 5: C/C++ program to identify valid keywords.

Output



```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

Active code page: 65001

D:\code2026\cse324>cd "d:\code2026\cse324\lab-mid\" && g++ 05-identify-valid-keywords.cpp -keywords && "d:\code2026\cse324\lab-mid\05-identify-valid-keywords
Enter a word: int
This is a Keyword.
d:\code2026\cse324\lab-mid>

```

Figure 5: Output of valid keyword identification program

6 Recognize Valid and Invalid Identifier

6.0.1 Question

1. Write a C/C++ program to recognize valid and invalid identifiers.

6.0.2 Solution

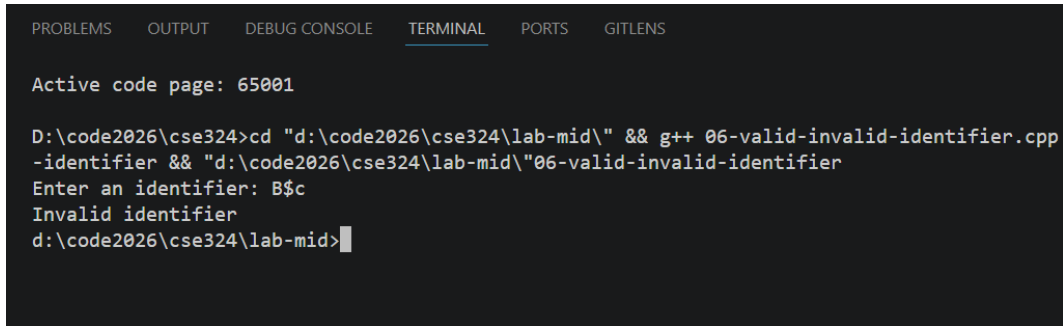
```

1 // Write a C/C++ program to recognize valid and invalid identifier.
2 #include <iostream>
3 #include <string>
4 using namespace std;
5
6 int main()
7 {
8     string id;
9     int valid = 1;
10
11     cout << "Enter an identifier: ";
12     cin >> id;
13
14     if (!((id[0] >= 'A' && id[0] <= 'Z') ||
15         (id[0] >= 'a' && id[0] <= 'z') ||
16         id[0] == '_'))
17     {
18         valid = 0;
19     }
20
21     for (int i = 1; i < id.length(); i++)
22     {
23         if (!((id[i] >= 'A' && id[i] <= 'Z') ||
24             (id[i] >= 'a' && id[i] <= 'z') ||
25             (id[i] >= '0' && id[i] <= '9') ||
26             id[i] == '_'))
27         {
28             valid = 0;
29             break;
30         }
31     }
32
33     if (valid == 1)
34         cout << "Valid identifier";
35     else
36         cout << "Invalid identifier";
37
38     return 0;
39 }

```

Listing 6: C/C++ program to recognize valid and invalid identifiers.

Output



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

Active code page: 65001

D:\code2026\cse324>cd "d:\code2026\cse324\lab-mid\" && g++ 06-valid-invalid-identifier.cpp
-invalid-identifier && "d:\code2026\cse324\lab-mid\"06-valid-invalid-identifier
Enter an identifier: B$c
Invalid identifier
d:\code2026\cse324\lab-mid>
```

Figure 6: Output of valid and invalid identifier recognition program

7 FIRST and FOLLOW Function

7.0.1 Question

1. Write a C/C++ program to determine the FIRST() and FOLLOW() function of the given grammar.

7.0.2 Solution

```
1  #include <iostream>
2  #include <string>
3  using namespace std;
4
5  string p[20];
6  int n;
7  char result[20];
8  int m = 0;
9
10 void add(char c)
11 {
12     for (int i = 0; i < m; i++)
13     {
14         if (result[i] == c)
15             return;
16     }
17
18     result[m++] = c;
19 }
20
21 int isNonTerminal(char c)
22 {
23     if (c >= 'A' && c <= 'Z')
24         return 1;
```

```

25     return 0;
26 }
27
28 void first(char c)
29 {
30     for (int i = 0; i < n; i++)
31     {
32         if (p[i][0] == c)
33         {
34             char x = p[i][2];
35
36             if (!isNonTerminal(x))
37             {
38                 add(x);
39             }
40             else
41             {
42                 first(x);
43             }
44         }
45     }
46 }
47
48 void follow(char c)
49 {
50     if (p[0][0] == c)
51         add('$');
52
53     for (int i = 0; i < n; i++)
54     {
55         for (int j = 2; j < p[i].length(); j++)
56         {
57             if (p[i][j] == c)
58             {
59                 if (j + 1 < p[i].length())
60                 {
61                     char next = p[i][j + 1];
62
63                     if (!isNonTerminal(next))
64                     {
65                         add(next);
66                     }
67                     else
68                     {
69                         first(next);
70                     }
71                 }
72                 else
73                 {
74                     if (p[i][0] != c)

```

```

75         follow(p[i][0]);
76     }
77 }
78 }
79 }
80 }
81
82 int main()
83 {
84     char ch;
85
86     cout << "Enter number of productions: ";
87     cin >> n;
88
89     cout << "Enter productions:" << endl;
90
91     for (int i = 0; i < n; i++)
92     {
93         cin >> p[i];
94     }
95
96     cout << "Enter variable: ";
97     cin >> ch;
98
99     m = 0;
100    first(ch);
101
102    cout << "FIRST(" << ch << ") = { ";
103    for (int i = 0; i < m; i++)
104        cout << result[i] << " ";
105    cout << "}" << endl;
106
107    m = 0;
108    follow(ch);
109
110    cout << "FOLLOW(" << ch << ") = { ";
111    for (int i = 0; i < m; i++)
112        cout << result[i] << " ";
113    cout << "}" << endl;
114
115    return 0;
116 }

```

Listing 7: C/C++ program to determine the FIRST() and FOLLOW() function of the given grammar.

Output



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

Active code page: 65001

D:\code2026\cse324>cd "d:\code2026\cse324\lab-mid\" && g++ 07-first-and-follow.cpp
d:\code2026\cse324\lab-mid\07-first-and-follow
Enter number of productions: 3
//Use # for epsilon
S=AB
A=a
B=b
Enter variable: A
FIRST(A) = { a }
FOLLOW(A) = { b }

d:\code2026\cse324\lab-mid>
```

Figure 7: Output of FIRST and FOLLOW function program

1 2

¹Kaniz Fatema

²ID: 20245103154